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Corrigendum

Corrigendum to “Grassmann phase space methods for fermions. II. Field theory” [Ann. Physics 377 (2017) 268–310]

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Errors occurred in SubSection 5.1 dealing with the application of the general Grassmann phase space field theory approach to a zero range two component model for a trapped spin 1/2 fermion gas. The error was due to not expressing the fermion–fermion interaction term (75) in the functional Fokker–Planck equation in the double space integral form (31) required for the general theory. From (31) the form of the diffusion matrix is identified, and then using Takagi factorisation as in (56), the noise term in the Ito stochastic field equation (51) can be obtained. This incorrect identification of the diffusion matrix affected SubSubSections 5.1.3 (Ito stochastic field equations) and 5.1.4 (Case of free Fermi gas). The corrected SubSubSections are set out below. The equation numbering is unchanged, but there are two additional un-numbered equations specifying the diffusion matrix.

5.1.3. Ito stochastic field equations

To determine the Ito stochastic field equations in the form (51) we must first put the fermion–fermion interaction term (75) in the form of a double space integral, so that the functional Fokker–Planck equation has the form given in (31). This can be done by introducing into (75) a double space integral $\iint dr ds$ along with a delta function $\delta(r - s) = \frac{1}{V} \sum_q \exp i q(r - s)$ expressed in terms of box normalised plane waves, with $V = L^3$ and (for 3D) $q \equiv \{q_x(\frac{2\pi}{L}), q_y(\frac{2\pi}{L}), q_z(\frac{2\pi}{L})\}$ -

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where q_x, q_y, q_z are all integers. In this case the non-zero elements of the diffusion matrix analogous to (31) are given by

$$\begin{aligned} D_{u1,d1}(\psi(r), \psi(s)) &= \frac{ig}{\hbar V} \sum_q \theta_{d1}^q(r) \theta_{u1}^q(s) & D_{d1,u1}(\psi(r), \psi(s)) &= \frac{ig}{\hbar V} \sum_q \theta_{u1}^q(r) \theta_{d1}^q(s) \\ D_{u2,d2}(\psi(r), \psi(s)) &= \frac{ig}{\hbar V} \sum_q \theta_{d2}^q(r) \theta_{u2}^q(s) & D_{d2,u2}(\psi(r), \psi(s)) &= \frac{ig}{\hbar V} \sum_q \theta_{u2}^q(r) \theta_{d2}^q(s) \end{aligned}$$

where

$$\begin{aligned} \theta_{d1}^q(r) &= \psi_{d1}(r) \exp(+iqr) & \theta_{u1}^q(r) &= \psi_{u1}(r) \exp(-iqr) \\ \theta_{d2}^q(r) &= \psi_{d2}(r) \exp(-iqr) & \theta_{u2}^q(r) &= \psi_{u1}(r) \exp(+iqr) \end{aligned}$$

The diffusion matrix can be factorised in the Takagi form [24] as in Eq. (56). After minor algebra the Ito field equations can be put in the form analogous to Eq (51) (see also Eq. (94) in Appendix E for the intermediate step). We have

$$\begin{aligned} & \begin{bmatrix} \tilde{\psi}_u(s, t + \delta t) \\ \tilde{\psi}_d(s, t + \delta t) \\ \tilde{\psi}_u^+(s, t + \delta t) \\ \tilde{\psi}_d^+(s, t + \delta t) \end{bmatrix} \\ &= \begin{bmatrix} \Theta_{u,u}(t^+) & \Theta_{u,d}(t^+) & 0 & 0 \\ \Theta_{d,u}(t^+) & \Theta_{d,d}(t^+) & 0 & 0 \\ 0 & 0 & \Theta_{u,u}^+(t^+) & \Theta_{u,d}^+(t^+) \\ 0 & 0 & \Theta_{d,u}^+(t^+) & \Theta_{d,d}^+(t^+) \end{bmatrix} \begin{bmatrix} \tilde{\psi}_u(s, t) \\ \tilde{\psi}_d(s, t) \\ \tilde{\psi}_u^+(s, t) \\ \tilde{\psi}_d^+(s, t) \end{bmatrix} \end{aligned} \quad (76)$$

where the Θ quantities are

$$\begin{aligned} [\Theta(t^+)] &= \begin{bmatrix} 1 - \frac{i}{\hbar} \left\{ -\frac{\hbar^2}{2m} \nabla^2 + V_u \right\} \delta t & \lambda \sum_q \exp(+iqs) \{ \delta \tilde{\omega}_{u,d}^q + i \delta \tilde{\omega}_{d,u}^q \} \\ \lambda \sum_q \exp(-iqs) \{ \delta \tilde{\omega}_{u,d}^q - i \delta \tilde{\omega}_{d,u}^q \} & 1 - \frac{i}{\hbar} \left\{ -\frac{\hbar^2}{2m} \nabla^2 + V_d \right\} \delta t \end{bmatrix} \\ [\Theta^+(t^+)] &= \begin{bmatrix} 1 + \frac{i}{\hbar} \left\{ -\frac{\hbar^2}{2m} \nabla^2 + V_u \right\} \delta t & \lambda \sum_q \exp(-iqs) \{ \delta \tilde{\omega}_{u+,d+}^q + i \delta \tilde{\omega}_{d+,u+}^q \} \\ \lambda \sum_q \exp(+iqs) \{ -\delta \tilde{\omega}_{u+,d+}^q + i \delta \tilde{\omega}_{d+,u+}^q \} & 1 + \frac{i}{\hbar} \left\{ -\frac{\hbar^2}{2m} \nabla^2 + V_d \right\} \delta t \end{bmatrix} \end{aligned} \quad (77)$$

where $\lambda = \sqrt{\frac{ig}{2\hbar V}}$. These quantities involve the Laplacian, the trap potentials, the time interval δt as well as four stochastic Wiener increments (68) $\delta \tilde{\omega}_{u,d}^q, \delta \tilde{\omega}_{d,u}^q, \delta \tilde{\omega}_{u+,d+}^q$ and $\tilde{\omega}_{d+,u+}^q$ for each q . The latter satisfy conditions analogous to (69). Note the sum over q .

This form of the Ito stochastic field equations shows that the Grassmann stochastic fields at time $t + \delta t$ are linearly related to the Grassmann stochastic fields at time t via quantities that only involve c-numbers. The quantities involve stochastic Wiener increments as well as the Laplacian and the trap potentials, but nevertheless no Grassmann variables are involved. A similar Grassmann independent feature applied to the situation where separate modes were treated (see Eq. (81) in Ref. [1]), and is a feature restricted to treatments involving the B distribution, as Plimak et al. first pointed out [8].

A key point to note is that there is no change in the spatial coordinates when the stochastic fields at time t are mapped onto the stochastic fields at time $t + \delta t$. This is a consequence of the zero range fermion-fermion interaction involved in the Hamiltonian (71).

The result in (76) (analogous to Eq. (81) for separate modes in Ref. [1]) shows how Grassmann stochastic fields at any time t_f can be linearly related to the Grassmann stochastic fields at an initial time t_0 via c-number stochastic quantities. Using the non-correlation result (70) the stochastic averages of products of Grassmann fields at any time t_f can be linearly related to the stochastic average of the products of the same number of Grassmann stochastic fields at an initial time t_0 via c-number stochastic quantities. The stochastic average of the products of the Grassmann stochastic fields at an initial time t_0 is determined from the initial conditions for the density operator.

5.1.4. Case of free Fermi gas

In the case of the free Fermi gas where the trap potentials are zero, a useful development of the result (76) can be obtained via introducing stochastic momentum fields via discrete Fourier transforms. The stochastic momentum fields $\tilde{\phi}_\alpha(k, t)$, $\tilde{\phi}_\alpha^\dagger(k, t)$ for wave number k are defined via

$$\begin{aligned}\tilde{\psi}_\alpha(s, t) &= \frac{1}{\sqrt{V}} \sum_k \exp(+iks) \tilde{\phi}_\alpha(k, t) \quad \alpha = u, d \\ \tilde{\psi}_\alpha^\dagger(s, t) &= \frac{1}{\sqrt{V}} \sum_k \exp(-iks) \tilde{\phi}_\alpha^\dagger(k, t) \quad \alpha = u, d\end{aligned}\quad (78)$$

together with the inverse relations. Here box normalisation has been used, where (for 3D) the wave number is quantised as $k \equiv \{k_x(\frac{2\pi}{L}), k_y(\frac{2\pi}{L}), k_z(\frac{2\pi}{L})\}$ - and k_x, k_y, k_z are all integers. The equations for $\tilde{\psi}_\alpha(k, t)$, $\tilde{\psi}_\alpha^\dagger(k, t)$ give stochastic equations for the momentum fields $\tilde{\phi}_\alpha(k, t)$, $\tilde{\phi}_\alpha^\dagger(k, t)$. We have

$$\begin{aligned}& \begin{bmatrix} \tilde{\phi}_u(k, t + \delta t) \\ \tilde{\phi}_d(k, t + \delta t) \\ \tilde{\phi}_u^\dagger(k, t + \delta t) \\ \tilde{\phi}_d^\dagger(k, t + \delta t) \end{bmatrix} \\ &= \sum_q \begin{bmatrix} F_{u,u}(q, t^+) & F_{u,d}(q, t^+) & 0 & 0 \\ F_{d,u}(q, t^+) & F_{d,d}(q, t^+) & 0 & 0 \\ 0 & 0 & F_{u,u}^+(q, t^+) & F_{u,d}^+(q, t^+) \\ 0 & 0 & F_{d,u}^+(q, t^+) & F_{d,d}^+(q, t^+) \end{bmatrix} \begin{bmatrix} \tilde{\phi}_u(k+q, t) \\ \tilde{\phi}_d(k-q, t) \\ \tilde{\phi}_u^\dagger(k-q, t) \\ \tilde{\phi}_d^\dagger(k+q, t) \end{bmatrix}\end{aligned}\quad (79)$$

where the F quantities are

$$\begin{aligned}[F(q, t^+)] &= \begin{bmatrix} \delta_{q,0} \left(1 - \frac{i}{\hbar} \left\{ \frac{\hbar^2 k^2}{2m} \right\} \delta t\right) & \lambda \{ \delta \tilde{\omega}_{u,d}^q + i \delta \tilde{\omega}_{d,u}^q \} \\ \lambda \{ \delta \tilde{\omega}_{u,d}^q - i \delta \tilde{\omega}_{d,u}^q \} & \delta_{q,0} \left(1 - \frac{i}{\hbar} \left\{ \frac{\hbar^2 k^2}{2m} \right\} \delta t\right) \end{bmatrix} \\ [F^+(q, t^+)] &= \begin{bmatrix} \delta_{q,0} \left(1 + \frac{i}{\hbar} \left\{ \frac{\hbar^2 k^2}{2m} \right\} \delta t\right) & \lambda \{ \delta \tilde{\omega}_{u+,d+}^q + i \delta \tilde{\omega}_{d+,u+}^q \} \\ \lambda \{ -\delta \tilde{\omega}_{u+,d+}^q + i \delta \tilde{\omega}_{d+,u+}^q \} & \delta_{q,0} \left(1 + \frac{i}{\hbar} \left\{ \frac{\hbar^2 k^2}{2m} \right\} \delta t\right) \end{bmatrix}\end{aligned}\quad (80)$$

which although still stochastic no longer involve the Laplacian operator – this has been replaced by the c-number k^2 . Note again the sum over q . These equations can be solved numerically. The stochastic averages of products of Grassmann stochastic momentum fields at a final time t can be linearly related to the stochastic average of the products of the same number of Grassmann stochastic momentum fields at an initial time t_0 via c-number stochastic quantities.

A key point to note is that there is now a change in the momentum coordinates $\hbar k$ when the stochastic fields at time t are mapped onto the stochastic fields at time $t + \delta t$. We see that $\tilde{\phi}_u(k, t + \delta t)$, $\tilde{\phi}_d(k, t + \delta t)$ not only depend on $\tilde{\phi}_u(k, t)$, $\tilde{\phi}_d(k, t)$ but also on $\tilde{\phi}_u(k+q, t)$, $\tilde{\phi}_d(k-q, t)$. Similar remarks apply to $\tilde{\phi}_u^\dagger(k, t + \delta t)$, $\tilde{\phi}_d^\dagger(k, t + \delta t)$.

The field operators $\hat{\psi}_\alpha(r)$, $\hat{\psi}_\alpha^\dagger(r)$ may be replaced by momentum field operators $\hat{\phi}_\alpha(k)$, $\hat{\phi}_\alpha^\dagger(k)$ via equations analogous to (78). The creation field momentum operator $\hat{\phi}_\alpha^\dagger(k)$ creates a fermionic atom

with spin α and with momentum $\hbar k$ and these are analogous to the mode creation operators $\hat{c}_{k,\alpha}^\dagger$ discussed in Section 6 of Paper 1 [1]. Note that both the Hamiltonian considered for the simple Cooper pair model in Section 6 of Ref. [1] and that in Eq. (71) are spin conserving and are the same. Application of the approach in the present section to the simple four mode model in Section 6 of Ref. [1] yields the same results as before for the coherence between momentum Fock states, which provides a check on Eqs. (79), (80).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.